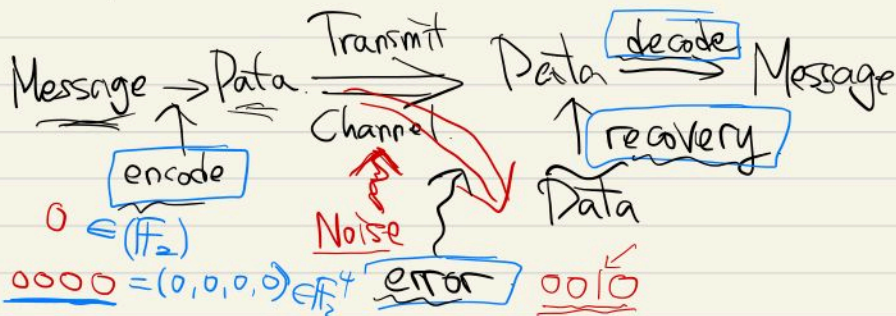


Quantum error correcting Code.

Alice

Bob.



Classical Error Correction.

1. Encode: $(\mathbb{F}_2)^K \xrightarrow{C_c} (\mathbb{F}_2)^n$ $K < n$
 $w \mapsto \mathbb{G}w$ $[C_c] = G$

Def (distance & weight). $u, v \in (\mathbb{F}_2)^n$.

$d_H(u, v) = \sum_{i=1}^n (u_i \oplus v_i)$ $\leftarrow u = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} \quad v = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$

$w_H(u) = d_H(u, 0) = \sum_{i=1}^n u_i$

Ex $d_H(001, 101) = 1 + 0 + 0 = 1$

$w_H(101) = 2$

$d_H(C_c) = \min \{ w_H(u) : u \in \text{im } C_c, u \neq 0 \}$

$$2. w \xrightarrow{C_c} \underset{\substack{\text{FW} \\ \parallel \\ C}}{\mathbb{F}_2^n} \implies \hat{c} = c + \underbrace{\epsilon}_{\text{error}}$$

3. Detect error & Recover.

$$(\mathbb{F}_2)^k \xrightarrow[G]{C_c} (\mathbb{F}_2)^n \quad \text{rk}(C_c) = k$$

Def (Parity Check Matrix).

A matrix $H \in M_{(n-k) \times n}(\mathbb{F}_2)$ s.t. $\ker H = \text{im}(C_c)$
 is called a parity check matrix of the code.

(of the encoding situation)

• H exists but NOT unique.

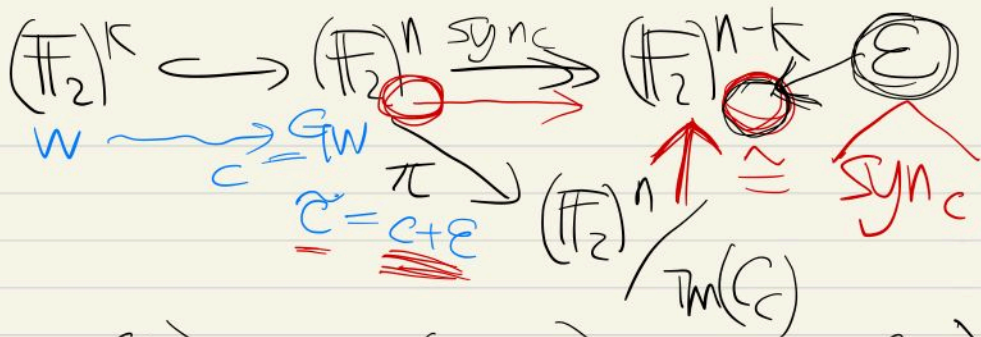
$$0 \rightarrow (\mathbb{F}_2)^k \xrightarrow[G]{C_c} (\mathbb{F}_2)^n \xrightarrow{H} (\mathbb{F}_2)^n / \text{im}(C_c) \rightarrow 0$$

$$\cong (\mathbb{F}_2)^{n-k}$$

Def (Syndrome Mapping).

$(\mathbb{F}_2)^n \xrightarrow{\text{syn}_C} (\mathbb{F}_2)^{n-k}$ is called a syndrome map.

The map



$$\text{syn}(\tilde{c}) = \text{syn}(c + e) = \text{syn}(e).$$

$$\text{Recover: } \tilde{c} + e = c \oplus e \oplus e = c$$

Thm (For "small" errors, we can recover correctly).

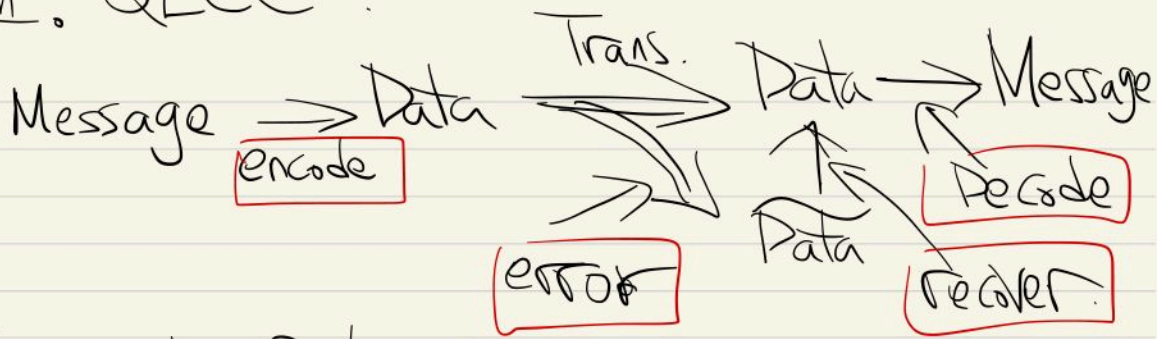
$$w_H(e) \leq \left\lfloor \frac{d_H(c) - 1}{2} \right\rfloor$$

$\lfloor 2.314 \rfloor = 2$
 $\lfloor 1.999 \rfloor = 1$

C_c : repetition

0 0000000

II. QECC



1. Encoding & decoding

Def ① An encoding is an injective & norm-preserving operation

$$C_g: \mathcal{H}^{\otimes K} \rightarrow \mathcal{H}^{\otimes n} \quad (K < n)$$

$$\mathcal{H}^{\otimes K} = (\mathbb{C}^2)^{\otimes K}, \quad \mathcal{H} = \mathbb{C}^2$$

$$\text{Im}(C_g) = \mathcal{H}^{C_g} \cdot D_g(x\rho + (1-x)\eta) = xD_g(\rho) + (1-x)D_g(\eta)$$

② Decoding is a convex-linear map

$$D_g: \mathcal{D}(\mathcal{H}^{C_g}) \rightarrow \mathcal{D}(\mathcal{H}^K)$$

satisfying $D_g(\underbrace{|\bar{\psi}\rangle\langle\bar{\psi}|}_{\in \mathcal{H}^{C_g}}) = |\bar{\psi}\rangle\langle\bar{\psi}|$

$\forall |\bar{\psi}\rangle \in \mathcal{H}^{C_g}$

$$e: \mathcal{H}^{C_g} \rightarrow \mathcal{H}^{C_g}$$

$$① e = e^*$$

$$② e \geq 0$$

$$③ \text{tr}(e) = 1$$

$$\eta \geq 0 \in \mathcal{H}$$

$$\downarrow$$

$$|\eta\rangle\langle\eta| \in \mathcal{D}(\mathcal{H})$$

$$||\cdot||$$

• $|x\rangle \in \mathcal{H}^k$. e.g. $k=3$. $|x\rangle = |001\rangle$
 $\mathcal{H}^k \xrightarrow{C_g} \mathcal{H}^n$
 $|x\rangle \mapsto |\underline{\overline{Tx}}\rangle = \underline{C_g|x\rangle}$
 $= |0\rangle \otimes |0\rangle \otimes |1\rangle$
 $= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

2. Error Operation

Def (Quantum Operation) on $\mathcal{H}$ is

① $K: \mathcal{D}(\mathcal{H}) \rightarrow \mathcal{D}(\mathcal{H})$
 convex-linear map. $\text{tr}(p) \leq 1$

② can be represented by $K(p) = \sum_{\ell=1}^m K_{\ell} p K_{\ell}^*$
 where $K_{\ell} \in \mathcal{L}(\mathcal{H})$

③ $\sum K_{\ell}^* K_{\ell} \leq I$, but " $=$ " $\Leftrightarrow \text{tr}(K(p)) = \text{tr}(p) = 1$

Def (Error operation)

An error operation $\mathcal{E}: \mathcal{D}(\mathcal{H}^n) \rightarrow \mathcal{P}_{\leq}(\mathcal{H}^n)$

$\mathcal{E}(p) = \sum_{i=1}^n \mathcal{E}_i p \mathcal{E}_i^*$ $\mathcal{E}_i \in \mathcal{L}(\mathcal{H}^n)$

error operator

Or. $\mathcal{E}_i = \sqrt{p_i} \hat{\mathcal{E}}_i$

$\Rightarrow \mathcal{E}(p) = \sum_i p_i \hat{\mathcal{E}}_i p \hat{\mathcal{E}}_i^*$

Set $\hat{E}_0 = 1$

Example $\mathcal{H}^1 \xrightarrow{C_g} \mathcal{H}^{\otimes 3} \xrightarrow{A} \mathcal{H}^3$

$(\mathbb{C}^2)^{\otimes 3} \cong \mathbb{C}^8$

$C_g: \mathcal{H}^1 \xrightarrow{\sim} \mathcal{H}^3 \xrightarrow{A} \mathcal{H}^3$

$|1\rangle \mapsto |1\rangle \otimes |0\rangle \otimes |0\rangle$

$A = |1\rangle\langle 1| \otimes \sigma_x \otimes \sigma_x$
 $+ |0\rangle\langle 0| \otimes 1 \otimes 1$

$(\cdot)(\cdot): \mathbb{C}^2 \rightarrow \mathbb{C}^2$

$C_g = A \circ \tau$

$\{C_g|0\rangle = |000\rangle \left(\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \right)$
 $\{C_g|1\rangle = |111\rangle \left(\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right)$

$\mathcal{E}: D(\mathcal{H}^{\otimes 3}) \rightarrow D(\mathcal{H}^{\otimes 3})$

$\mathcal{E}_{00} = 1 \otimes 1 \otimes 1$ $\mathcal{E}_{01} = 1 \otimes 1 \otimes \sigma_x$

$\mathcal{E}_{10} = 1 \otimes \sigma_x \otimes 1$ $\mathcal{E}_{11} = \sigma_x \otimes 1 \otimes 1$

$\mathcal{E}_{01}|000\rangle = |001\rangle$

Pauli Matrix

$\sigma_0 = 1$

$\sigma_1 = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

$\sigma_2 = \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

$\sigma_3 = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

3. Recovery Operation

Def

$$\tilde{C} = C + E$$

$$\tilde{C} + \boxed{E} = C + E + E = C$$

① $R: D(\mathcal{H}^{\otimes n}) \rightarrow D(\mathcal{H}^{\otimes n})$ quantum operation which preserves the trace.

$$R(\rho) = \sum_i R_i \rho R_i^* \quad R_i \in \mathcal{L}(\mathcal{H}^{\otimes n})$$

↖ recovery operator

② We say an error operation E is correctable if $\exists$ recovery operation R s.t.

$$\forall \rho \in \underbrace{D(\mathcal{H}_{C_q})}_{\text{im}(C_q)}, \quad R\left(\frac{E(\rho)}{\text{tr}(E(\rho))}\right) = \rho.$$

$$E(\rho) \in \mathcal{D}(\mathcal{H}^n)$$

Theorems on determining "correctable"

$$\textcircled{1} \iff \text{tr}(E(\rho)) = \text{const } K_E \in [0,1] \quad \forall \rho.$$

Thm $E = \{E_i\}_{i=1}^n$ is correctable

$$\iff \exists C \in \text{Mat}_{n \times n}(\mathbb{C}) \text{ s.t. } C = C^*$$

② $\forall |\pi_x\rangle$ ONB of $\mathcal{H}_{C_q}$, we have

$$\langle E_i \pi_x | E_j \pi_y \rangle = C_{ij} \delta_{xy}$$

$$\mathcal{H}^{\otimes n} \xrightarrow{C_g} \mathcal{H}^{\otimes n}$$

$$\uparrow (C)_{ij}$$

$$|\alpha\rangle \mapsto |\mathbb{I}\alpha\rangle$$

ONB

Thm ("small" errors are correctable).

$$\mathcal{E} = \{\mathcal{E}_i\} \text{ } n\text{-qubit error.}$$

$$n \leq \left\lfloor \frac{d_p(C_g) - 1}{2} \right\rfloor. \text{ Then } \mathcal{E} \text{ is correctable.}$$

* $\mathcal{E} = \{\mathcal{E}_i\}$ is k-qubit if $\mathcal{E}_i$ s are k-local.

$$\begin{aligned} \mathcal{E}_i &\in \mathcal{L}(\mathcal{H}^{\otimes n}) \cong \mathcal{L}(\mathcal{H}) \otimes \mathcal{L}(\mathcal{H}) \otimes \dots \otimes \mathcal{L}(\mathcal{H}) \\ &= \mathcal{E}_i = \sum a \underbrace{A_1 \otimes A_2 \otimes \dots \otimes A_n}_{k\text{-local}} \end{aligned}$$

$$|\{j \in \{1, 2, \dots, n\} : A_j \neq I\}| \leq k$$

e.g. $\mathcal{E} = \underbrace{\sigma_x \otimes \sigma_y \otimes I \otimes I}_{2\text{-local}} + \underbrace{\sigma_y \otimes I \otimes I \otimes \sigma_x}_{2\text{-local}}$

4. Detect & Correct. $\Rightarrow$ Syndrome.

Def $\mathcal{H}^k \xrightarrow{C_g} \mathcal{H}^n$ $\mathcal{E} = \{\mathcal{E}_i\}_{i \in I} \neq 0$
 also $\mathcal{E}_e = \sqrt{p_i} \mathcal{E}_i$, $\mathcal{E}_i \in \mathcal{U}(\mathcal{H}^n)$

Define $\mathcal{H}^S$ to be a Hilbert space of
 $\dim = |I|$. (i.e. $\mathcal{H}^S \cong \mathbb{C}^{|I|}$)
 and $\{|\varphi_i\rangle\}_{i \in I} = \text{ONB of } \mathcal{H}^S$.

Then the syndrome operator $S_{\mathcal{E}}$ is
 a norm-preserving operator

$S_{\mathcal{E}}: \mathcal{H}^n \rightarrow \mathcal{H}^n \otimes \mathcal{H}^S$ s.t.

$$S_{\mathcal{E}}(\mathcal{E}_a|\pi\rangle) = \mathcal{E}_a|\pi\rangle \otimes |\varphi_a\rangle$$

$\forall \mathcal{E}_a, \forall \pi \in \mathcal{H}^G$.

$a = \text{syn}_g(\mathcal{E}_a)$ syndrome of $\mathcal{E}_a$.

$S_{\mathcal{E}}$ (what he got but don't know) = $(\text{ }) \otimes |\varphi_{\mathcal{E}_3}\rangle$

Example $\mathcal{H}^1 \hookrightarrow \mathcal{H}^3 \quad C_g = A_{07}$

$$\left\{ \begin{array}{l} |0\rangle \mapsto |000\rangle \\ |1\rangle \mapsto |111\rangle \end{array} \right.$$

errors $E_{00} = | \otimes | \otimes | \quad E_{10} = | \otimes \sigma_x \otimes |$
 $E_{01} = | \otimes | \otimes \sigma_x \quad E_{11} = \sigma_x \otimes | \otimes |$

$$S_E : \mathcal{H}^{\otimes 3} \longrightarrow \mathcal{H}^{\otimes 3} \otimes \mathcal{H}^{\otimes 2} \cong \begin{pmatrix} \mathbb{C}^2 \otimes \mathbb{C}^2 \\ \cong \mathbb{C}^4 \end{pmatrix}$$

$$\begin{aligned} & (|111\rangle\langle 111| + |000\rangle\langle 000|) \otimes |00\rangle \\ & + (|110\rangle\langle 110| + |001\rangle\langle 001|) \otimes |01\rangle \\ & + (|101\rangle\langle 101| + |010\rangle\langle 010|) \otimes |10\rangle \\ & + (|100\rangle\langle 100| + |011\rangle\langle 011|) \otimes |11\rangle. \end{aligned}$$

$$|\Psi\rangle = a|000\rangle + b|111\rangle \in \mathcal{H}^{\otimes 3}$$

$$E_{10} |\Psi\rangle = a|010\rangle + b|101\rangle$$

$$S_E(E_{10} |\Psi\rangle) = (a|010\rangle + b|101\rangle)$$

$$\otimes |10\rangle$$

$$E_{10}$$

$$P_{10} = \mathcal{E}_{10}^*$$

$$\Rightarrow \mathcal{E}_{10}^* (\mathcal{E}_{10} |\overline{\Psi}\rangle) = \underbrace{\mathcal{E}_{10}^* \mathcal{E}_{10}}_{\mathbb{I}} |\overline{\Psi}\rangle$$

Fact list of Pauli Group.

• Pauli Group P_n .

1) $\sigma_0, \sigma_1, \sigma_2, \sigma_3 \in \mathcal{L}(\mathbb{C}^2)$. 2×2

$\sigma_x, \sigma_y, \sigma_z$

properties ① $\sigma^* = \sigma$, $\sigma \sigma^* = \sigma^* \sigma \Rightarrow \sigma^2 = \mathbb{I}$

② $\{\sigma_0, \sigma_1, \sigma_2, \sigma_3\}$ spans $\mathcal{L}(\mathbb{C}^2)$.

③ $\{i^\alpha \sigma_\beta\}_{\substack{\alpha=0,1,2,3 \\ \beta=0,1,2,3}}$ forms a group P_1

2) In general, $\mathcal{L}((\mathbb{C}^2)^{\otimes n})$

$\cong \mathcal{L}(\mathbb{C}^2) \otimes \mathcal{L}(\mathbb{C}^2) \otimes \dots$

$P_n = \{ \underline{i}^\alpha \sigma_{a_{n-1}} \otimes \sigma_{a_{n-2}} \otimes \dots \otimes \sigma_{a_0} \}$

forms a group.

$(\sigma_x \otimes \sigma_y) \cdot (\sigma_y \otimes \sigma_z) = \sigma_x \sigma_y \otimes \sigma_y \sigma_z$

$(2v \otimes w = v \otimes 2w)$

3) ① $\mathcal{P}_n$ "spans" $\mathcal{L}((\mathbb{C}^2)^{\otimes n})$
 $\forall T \in \mathcal{L}((\mathbb{C}^2)^{\otimes n}), \exists a_j \in \mathbb{C}, h_j \in \mathcal{P}_n$
 s.t. $T = \sum_{\text{finite}} a_j h_j$

② $\forall g, h \in \mathcal{P}_n, gh = hg$
 or $gh = -hg$

③ $\omega_p(\sigma_x \otimes \sigma_z \otimes 1) = 2$
 $\omega_p(\sigma_x \otimes \sigma_x \otimes \sigma_y \otimes 1 \otimes \sigma_z) = 4$

$d_p(\mathcal{C}_g) = \min \{ \omega_p(g) : g \in \mathcal{P}_n \text{ s.t. } \exists |x\rangle, |y\rangle$

$\in \text{ONB of } \mathcal{H}^{\mathcal{C}_g},$

satisfying $\langle x | g | y \rangle \neq f(g) \delta_{xy}$

$\forall f: \mathcal{P}_n \rightarrow \mathbb{C} \}$

$\boxed{\mathcal{P}_n \subset \mathcal{H}^n, \text{Stab}_{\mathcal{P}_n}(\mathcal{H}^{\mathcal{C}_g})}$